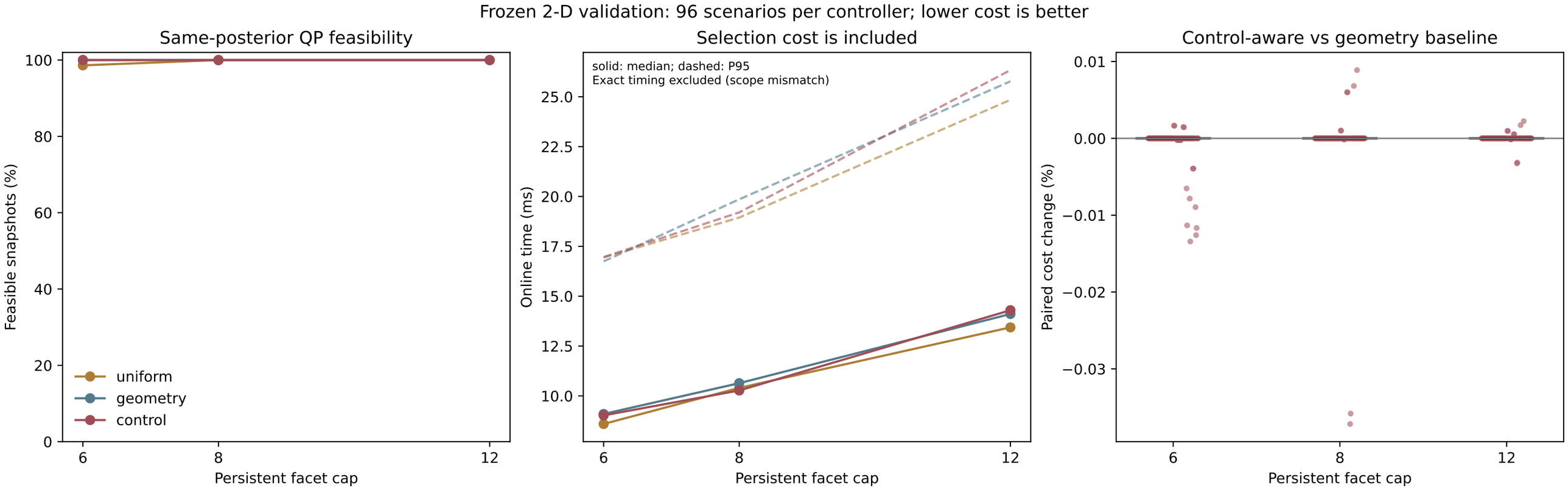
Frozen 2-D validation: 96 scenarios per controller; lower cost is better
Same-posterior QP feasibility
Feasible snapshots (%)
100
80
60
40
20
0
uniform
geometry
control
6
8
12
Persistent facet cap
Selection cost is included
solid: median; dashed: P95
Exact timing excluded (scope mismatch)
Online time (ms)
25.0
22.5
20.0
17.5
15.0
12.5
10.0
6
8
12
Persistent facet cap
Control-aware vs geometry baseline
Paired cost change (%)
0.01
0.00
−0.01
−0.02
−0.03
6
8
12
Persistent facet cap